Date:

24k - 0722 Abdul Rehman Asad

BCS - 5A 'CN'

Assignment - 2

Task 1:

1) Reliable data transfer is important because the network carries casualty lists, supply requests, and rescue assignments. If a packet is lost, corrupted or received out of order, the rescue team may receive incorrect information Therefore packets must be delivered correctly in order.

2) Stop & Wait sends only one packet & waits for its ACK before sending the next packet. In this scenario the RTT is 500ms, while transmitting an 8000-bit packet at 2MBps takes only 4ms. Therefore, the sender spends most of its time waiting instead of transmitting data.

3) Pipelining allows the sender to transmit multiple packets before receiving their ACKs. This reduces waiting time and improves link utilization It is useful on this satellite link because the RTT is much larger than the packet transmission time.

4) Selective Repeat (SR) is more suitable for this scenario for two reasons. First it re-transmits only the lost or corrupted packets instead of retransmitting all outstanding packets. Second its receiver can buffer out-of-order packets and deliver them in order when the missing packets arrive, which reduces unnecessary re-transmissions on the limited bandwidth satellite link.

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Go-Back-N (GBN)

Event	Sender	Packet	Receiver	ACk	Sender window
1	Send Pkt 0	0	Accept pkt 0	ACk 0	[0-3]
2	Send Pkt 1	1	Accept pkt 1	ACk 1	[0-3]
3	Send pkt 2	2	Packet loss	-	[0-3]
4	Send Pkt 3	3	Discard. out of order	ACk -1	[0-3]
5	Receive ACk 0, send 4	1	Discard if it arrives before pkt 2	ACk -1	[1-4]
6	Rec ACk 1; send pkt 5	5	Discard if it arrives before pkt 2	ACk -1	[2-5]
7	Timeout for pkt 2	2	-	-	[2-5]
8	Retransmit pkt 2	2	Accept pkt 2	ACk 2	[2-5]
9	Retransmit pkt 3	3	Accept pkt 3	ACk 3	[3-6]
10	Retransmit pkt 4	4	Accept pkt 4	ACk 4	[4-7]
11	Retransmit pkt 5	5	Accept pkt 5	ACk 5	[5-8]
12	Send remaining pkt as window advances	6-9	Accept in order	ACk 6-9	advances until complete.

Selective Repeat (SR):

Event	Sender	Packet	Receiver	ACk	Sender window
1	Send pkt 0	0	Accept & deliver pkt 0	ACk 0	[0-3]
2	Send pkt 1	1	Accept & deliver pkt 1	ACk 1	[0-3]
3	Send pkt 2	2	Packet loss	-	[0-3]
4	Send pkt 3	3	Buffer pkt 3	ACk 3	[0-3]
5	Rec ACk 0, send pkt 4	4	Buff pkt 4; bc pkt 2 still missing	ACk 4	[1-4]
6	Rec ACk 1, send pkt 5	5	Buff pkt 5 & forwards within	ACk 5	[2-5]
7	Timeout for pkt 2	2	-	-	[2-5]
8	Retransmit pkt 2 only	2	Accept pkt 2; deliver buffered pkt in order	ACk 2	ACks are processed
9	Process ACKs for buffered pkt	3-5	Deliver any remaining but pkts	individual ACKs	windows advances
10	Send remaining pkts as window advances	6-9	Accept & deliver in order	ACk 6-9	advances until complete.

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Comparision:

GBN: It retransmit pkt 2 & all other outstanding pkts in the sender window that have not been cumulatively ack In the trace above, pkt 3, 4, & 5 includes,

SR: It retransmits only pkt 2, assuming the other pkts & their ACKs correct.

Why Diff: GBN discards out-of-order packets, while SR buffers them & acknowledges them individually.

Task-5

1. Packet size, $L = 8000$ bits

Link rate, $R = 2$ Mbps $= 2000\,000$ bits/s

$RTT = 500$ ms $= 0.5$ s

Window size, $N = 4$

$\frac{L}{R} = \frac{8000}{2000\,000} = 0.004$ s

$\Rightarrow U = \frac{L/R}{RTT + L/R} = \frac{0.004}{0.5 + 0.004} = 0.79\%$

Pipelining with $N = 4$

$U = \min\left(1, \frac{N(L/R)}{RTT + L/R}\right) = \frac{4(0.004)}{0.504} = 3.17\%$

2) a) $k = 3$ bits

$2^k = 2^3 = \boxed{8}$

Sequence No = 0, 1, 2, 3, 4, 5, 6, and 7

b) For selective repeat, max window size is half of seq

$N_{max} = \frac{2^k}{2} = \frac{8}{2} = \boxed{4}$

c) A larger window can overlap with sequence numbers from an earlier transmission after the sequence numbers wrap around. The receiver may mistake an old duplicate packet for a new packet with the same sequence numbers.

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Example of an unsafe SR window size of 5

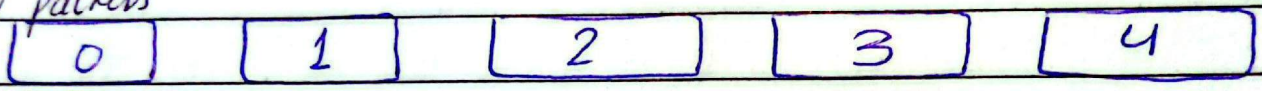
Sequence-number wraparound

Earlier Packets



after sequence no 7, numbering wraps back to 0

New packets



Problem- an old duplicate with sequence no 0 may be confused with a new packet numbered 0 if receiver's window overlaps.

* Therefore, SR window must be no larger than 4 using 3-bit sequence numbers.